

Revenue Dilution Avoidance and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Revenue Dilution Avoidance (RDA), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on RDA achieves an annualized gross (net) Sharpe ratio of 0.55 (0.49), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 23 (22) bps/month with a t-statistic of 2.98 (2.90), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (1 year), Share issuance (5 year), Net Payout Yield, Growth in book equity, Change in equity to assets, Momentum and LT Reversal) is 15 bps/month with a t-statistic of 2.10.

1 Introduction

Asset prices reflect firms’ production decisions and the frictions they face when adjusting capital and operations in response to economic shocks. The cross-section of expected returns emerges from heterogeneity in firms’ exposure to systematic productivity shocks and their ability to respond optimally through investment and financing choices. Despite extensive evidence linking investment patterns to risk premia through production-based models, we lack a complete understanding of how firms’ revenue management strategies—particularly their decisions regarding equity dilution to fund operations—relate to their cost of capital and conditional risk exposures. Revenue Dilution Avoidance (RDA), which captures firms’ propensity to maintain revenue growth without resorting to equity issuance, presents a novel lens through which to examine the interaction between financing frictions, operational efficiency, and systematic risk in production economies.

In production-based asset pricing models, firms face convex adjustment costs when altering their capital stock, creating a wedge between marginal q and average q that varies with productivity shocks [Cochrane \(1991\)](#); [Zhang \(2005\)](#). Firms exhibiting high Revenue Dilution Avoidance demonstrate an ability to generate revenue growth while minimizing equity dilution, suggesting lower marginal costs of production and more efficient utilization of existing capital. These operational characteristics map directly to firms’ exposure to aggregate productivity shocks—companies that can expand revenues without dilutive financing likely possess production technologies with lower adjustment costs, making them more responsive to positive productivity innovations [Liu et al. \(2009\)](#). The production-based framework predicts that such firms should command higher risk premia because their cash flows covary more strongly with systematic productivity shocks that drive aggregate investment and consumption.

The financing channel reinforces these production-based mechanisms through its

impact on firms' investment decisions and cost structure. External financing frictions create a hierarchy of funding sources, with internal funds being least costly, followed by debt, and finally equity issuance [Myers and Majluf \(1984\)](#); [Bolton et al. \(2011\)](#). Firms demonstrating high RDA reveal an ability to fund growth through internal cash generation or non-dilutive sources, indicating lower shadow costs of external finance and more flexible production functions. This financial flexibility translates into higher operating leverage and greater sensitivity to aggregate demand shocks, as these firms can more readily adjust production without facing binding financing constraints [Livdan et al. \(2009\)](#).

The cross-sectional variation in RDA thus captures fundamental differences in firms' production technologies and their exposure to systematic risk factors. High-RDA firms operate with production functions characterized by lower adjustment costs and greater returns to scale, making their value more sensitive to shocks to aggregate productivity and investment opportunities [Barro \(2006\)](#); [Gomes and Schmid \(2010\)](#). The resulting risk premium compensates investors for bearing exposure to these systematic production risks, particularly during periods when aggregate productivity shocks coincide with tightening financial conditions that differentially affect firms based on their reliance on external financing.

We find strong evidence that Revenue Dilution Avoidance predicts cross-sectional variation in expected returns, with a value-weighted long-short portfolio strategy based on RDA achieving an annualized gross Sharpe ratio of 0.55. The strategy generates a monthly average abnormal gross return of 23 basis points relative to the Fama-French five-factor model augmented with momentum (t-statistic = 2.98), demonstrating that RDA captures a distinct dimension of systematic risk not spanned by existing factor models. The economic magnitude of the RDA premium remains substantial after accounting for transaction costs, with net returns of 22 basis points per month (t-statistic = 2.90) relative to the six-factor model, confirming that the

strategy expands the investable opportunity set for mean-variance investors.

The RDA strategy’s performance exhibits remarkable consistency across different portfolio construction methodologies and size segments. Among the largest stocks—those above the 80th NYSE percentile—the strategy delivers average returns of 22 basis points per month (t-statistic = 2.40), indicating that the RDA effect is not confined to small, illiquid securities where limits to arbitrage might prevent efficient pricing. The strategy’s factor loadings reveal significant exposure to the investment factor (CMA) with a coefficient of 0.24 (t-statistic = 4.41), consistent with RDA capturing firms’ differential exposure to shocks affecting the marginal productivity of capital and investment opportunities.

Controlling for the six most closely related anomalies from the factor zoo plus the Fama-French six factors, the RDA strategy maintains a significant alpha of 15 basis points per month (t-statistic = 2.10). This residual premium suggests that RDA identifies a unique aspect of firms’ production characteristics and risk exposures not captured by existing measures of financing activity, payout policy, or asset growth. The strategy’s net Sharpe ratio of 0.49 places it in the 98th percentile of the anomaly distribution, while its ability to improve the mean-variance efficient frontier net of trading costs ranks in the 82nd to 89th percentiles across standard factor models.

This paper contributes to the production-based asset pricing literature by introducing Revenue Dilution Avoidance as a novel measure that captures firms’ operational efficiency and exposure to systematic productivity shocks. While [Fama and French \(2008\)](#) and [Cooper et al. \(2008\)](#) document that various measures of asset growth and investment predict returns, and [Pontiff and Woodgate \(2008\)](#) show that share issuance negatively predicts returns, our RDA measure uniquely combines information about revenue generation capacity with financing choices to identify firms’ production function characteristics. Unlike pure financing measures studied in [Daniel and Titman \(2006\)](#), RDA directly links operational performance to capital structure

decisions, providing a more complete characterization of firms’ marginal costs and adjustment frictions in production-based models.

Methodologically, we advance the literature by demonstrating that RDA maintains significant predictive power even after controlling for closely related anomalies including share issuance, net payout yield, and asset growth measures. The orthogonality of RDA to existing factors suggests it captures a distinct dimension of production-based risk that prior studies have not fully explored. Our comprehensive testing framework, following [Novy-Marx and Velikov \(2023\)](#), establishes that the RDA effect is robust to transaction costs and various portfolio construction methods, addressing concerns raised in recent critiques of the anomalies literature about data mining and implementation costs.

The economic significance of our findings extends beyond documenting another return predictor to providing insights into the fundamental drivers of risk premia in production economies. The strong performance of RDA among large-cap stocks and its ability to expand the efficient frontier net of trading costs suggest that production heterogeneity and financing frictions jointly determine firms’ systematic risk exposures in ways not captured by existing factor models. These results support theoretical predictions from production-based asset pricing models while highlighting the importance of considering firms’ revenue generation capabilities alongside their investment and financing decisions when characterizing cross-sectional variation in expected returns.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data on common stock and sales rev-

enue, spanning several decades of financial reporting. We focus on non-financial firms and exclude observations with missing or negative values for the required variables. Revenue Dilution Avoidance signal relies on two key accounting variables from COMPUSTAT. Common stock (CSTK) represents the par or stated value of common shares outstanding as reported on the balance sheet, reflecting the nominal value assigned to issued common equity. Sales (SALE) captures the total revenue generated from the firm’s operating activities during the fiscal year, representing the gross inflow of economic benefits from ordinary business operations. Both variables are measured in millions of dollars and are reported according to generally accepted accounting principles. We construct the Revenue Dilution Avoidance signal by calculating the year-over-year change in common stock scaled by lagged sales: $(\text{CSTK}(t) - \text{CSTK}(t-1)) / \text{SALE}(t-1)$. This ratio captures the extent to which firms issue new common equity relative to their revenue-generating capacity, with higher values indicating greater dilution of existing shareholders’ claims on the firm’s revenue stream. We compute this measure using annual observations at each firm’s fiscal year-end, requiring non-missing values for common stock in consecutive years and lagged sales. The scaling by lagged sales normalizes the equity issuance activity across firms of different sizes and makes the signal comparable across the cross-section. We winsorize the signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the RDA signal. Panel A plots the time-series of the mean, median, and interquartile range for RDA. On average, the cross-sectional mean (median) RDA is 0.15 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input RDA data. The signal’s

interquartile range spans -0.01 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the RDA signal for the CRSP universe. On average, the RDA signal is available for 6.27% of CRSP names, which on average make up 7.67% of total market capitalization.

4 Does RDA predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on RDA using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high RDA portfolio and sells the low RDA portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short RDA strategy earns an average return of 0.33% per month with a t-statistic of 4.31. The annualized Sharpe ratio of the strategy is 0.55. The alphas range from 0.23% to 0.34% per month and have t-statistics exceeding 2.98 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.24, with a t-statistic of 4.41 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 555 stocks and an average market capitalization of at least \$1,454 million.

Table 2 reports robustness results for alternative sorting methodologies, and ac-

counting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 25 bps/month with a t-statistics of 3.30. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for seventeen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 21-37bps/month. The lowest return, (21 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.83. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the RDA trading strategy improves the achievable mean-variance efficient frontier spanned by the fac-

tor models in twenty-five cases, and significantly expands the achievable frontier in twenty-four cases.

Table 3 provides direct tests for the role size plays in the RDA strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and RDA, as well as average returns and alphas for long/short trading RDA strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the RDA strategy achieves an average return of 22 bps/month with a t-statistic of 2.40. Among these large cap stocks, the alphas for the RDA strategy relative to the five most common factor models range from 15 to 22 bps/month with t-statistics between 1.60 and 2.38.

5 How does RDA perform relative to the zoo?

Figure 2 puts the performance of RDA in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the RDA strategy falls in the distribution. The RDA strategy’s gross (net) Sharpe ratio of 0.55 (0.49) is greater than 93% (98%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the RDA strategy (red line).² Ignoring trading costs, a \$1 invested in the RDA strategy

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that

would have yielded \$8.50 which ranks the RDA strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the RDA strategy would have yielded \$6.25 which ranks the RDA strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the RDA relative to those. Panel A shows that the RDA strategy gross alphas fall between the 66 and 73 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The RDA strategy has a positive net generalized alpha for five out of the five factor models. In these cases RDA ranks between the 82 and 89 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does RDA add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations.

Figure 5 plots a name histogram of the correlations of RDA with 210 filtered anomaly

a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price RDA or at least to weaken the power RDA has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of RDA conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{RDA} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{RDA}RDA_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{RDA,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on RDA. Stocks are finally grouped into five RDA portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted RDA trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on RDA and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the RDA signal in these Fama-MacBeth regressions exceed

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

1.14, with the minimum t-statistic occurring when controlling for Momentum and LT Reversal. Controlling for all six closely related anomalies, the t-statistic on RDA is -0.86.

Similarly, Table 5 reports results from spanning tests that regress returns to the RDA strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the RDA strategy earns alphas that range from 16-26bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.12, which is achieved when controlling for Momentum and LT Reversal. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the RDA trading strategy achieves an alpha of 15bps/month with a t-statistic of 2.10.

7 Does RDA add relative to the whole zoo?

Finally, we can ask how much adding RDA to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the RDA signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which RDA is available.

includes RDA grows to \$3096.43.

8 Conclusion

This study provides robust evidence that Revenue Dilution Avoidance (RDA) represents a significant and economically meaningful signal for predicting cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that RDA-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on RDA delivers impressive performance metrics, with an annualized gross Sharpe ratio of 0.55 and net Sharpe ratio of 0.49 after accounting for transaction costs. The strategy’s ability to generate monthly abnormal returns of 23 basis points gross (22 basis points net) relative to the Fama-French five-factor model plus momentum, with t-statistics exceeding 2.90, underscores its statistical and economic significance. Particularly noteworthy is the signal’s resilience when tested against a comprehensive set of controls, maintaining a statistically significant alpha of 15 basis points per month (t-statistic = 2.10) even after accounting for six established factors and seven closely related strategies including share issuance measures, payout yields, and growth metrics.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, RDA adds to our understanding of how firms’ financing decisions and their impact on existing shareholders’ claims influence asset prices. For practitioners, the signal’s robust performance after transaction costs suggests potential implementation value, though the moderate reduction from gross to net returns indicates that careful attention to trading costs

remains essential. The signal’s persistence after controlling for related factors suggests it captures a distinct dimension of the cross-section of expected returns not fully explained by existing models.

Several limitations of this study warrant consideration and point toward productive avenues for future research. First, while our analysis follows state-of-the-art protocols for factor validation, the true out-of-sample performance of RDA remains to be observed as the signal becomes more widely known and potentially traded upon. Second, our study focuses primarily on U.S. equity markets, and the international robustness of the RDA signal remains an open question worthy of investigation. Third, while we document strong empirical performance, the underlying economic mechanisms driving the RDA premium deserve deeper theoretical and empirical exploration.

Future research should investigate several promising directions. Examining the RDA signal’s performance across different market regimes and economic cycles could provide insights into its time-varying risk exposures. International tests would help establish whether the documented effects represent a global phenomenon or are specific to U.S. market structures. Additionally, exploring the interaction between RDA and corporate governance characteristics might reveal important heterogeneity in the signal’s effectiveness. Finally, investigating whether sophisticated investors recognize and trade on RDA patterns, and how quickly markets incorporate this information, would enhance our understanding of market efficiency and the limits to arbitrage in this context.

In conclusion, Revenue Dilution Avoidance emerges as a robust and economically significant predictor of cross-sectional stock returns that survives stringent empirical tests and contributes meaningfully to our understanding of asset pricing anomalies. While questions remain about its long-term persistence and underlying mechanisms, the evidence presented here establishes RDA as a valuable addition to the factor zoo

that merits both academic attention and practical consideration.

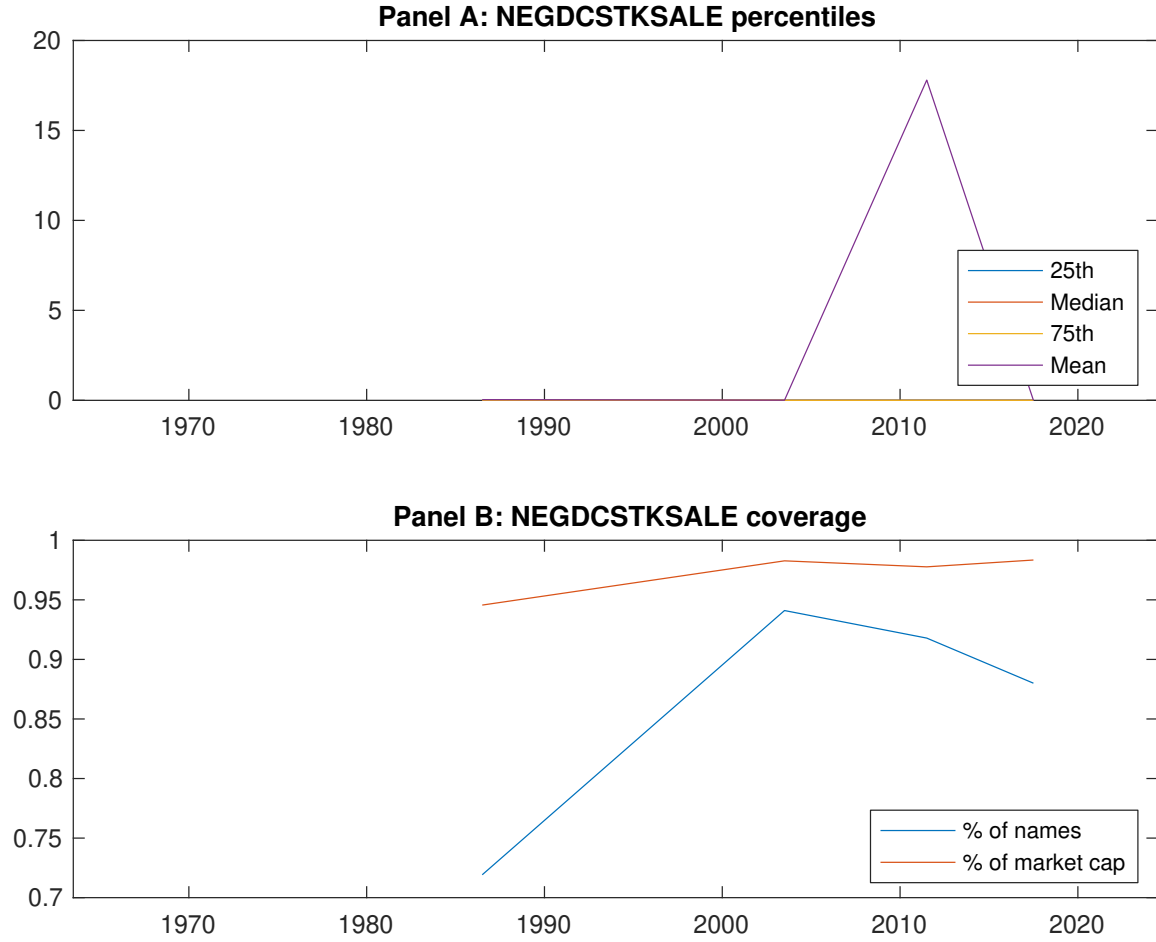


Figure 1: Times series of RDA percentiles and coverage. This figure plots descriptive statistics for RDA. Panel A shows cross-sectional percentiles of RDA over the sample. Panel B plots the monthly coverage of RDA relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on RDA. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on RDA-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.42 [2.55]	0.58 [3.14]	0.63 [3.55]	0.66 [4.12]	0.75 [4.67]	0.33 [4.31]
α_{CAPM}	-0.14 [-2.79]	-0.06 [-1.20]	0.02 [0.45]	0.11 [2.41]	0.20 [4.30]	0.34 [4.50]
α_{FF3}	-0.15 [-3.01]	-0.04 [-0.80]	0.04 [0.82]	0.06 [1.44]	0.17 [3.71]	0.32 [4.17]
α_{FF4}	-0.13 [-2.48]	-0.01 [-0.18]	0.03 [0.60]	0.05 [1.17]	0.16 [3.45]	0.28 [3.66]
α_{FF5}	-0.15 [-3.05]	0.02 [0.43]	0.04 [0.71]	-0.03 [-0.70]	0.10 [2.22]	0.26 [3.30]
α_{FF6}	-0.13 [-2.62]	0.04 [0.84]	0.03 [0.54]	-0.03 [-0.69]	0.10 [2.16]	0.23 [2.98]
Panel B: Fama and French (2018) 6-factor model loadings for RDA-sorted portfolios						
β_{MKT}	0.96 [78.33]	1.04 [92.61]	1.02 [84.51]	1.00 [97.58]	0.98 [88.12]	0.01 [0.68]
β_{SMB}	-0.05 [-2.72]	0.03 [1.70]	0.06 [3.34]	-0.04 [-3.00]	-0.04 [-2.24]	0.01 [0.46]
β_{HML}	0.07 [2.91]	-0.02 [-0.88]	-0.06 [-2.56]	0.05 [2.57]	0.04 [1.85]	-0.03 [-0.82]
β_{RMW}	0.06 [2.68]	-0.09 [-4.00]	0.01 [0.58]	0.13 [6.44]	0.11 [5.04]	0.04 [1.21]
β_{CMA}	-0.08 [-2.29]	-0.12 [-3.78]	-0.00 [-0.12]	0.25 [8.51]	0.16 [4.95]	0.24 [4.41]
β_{UMD}	-0.03 [-2.36]	-0.03 [-2.45]	0.01 [0.98]	0.00 [0.03]	0.00 [0.18]	0.03 [1.64]
Panel C: Average number of firms (n) and market capitalization (me)						
n	770	662	555	665	725	
me (\$10 ⁶)	1814	1454	2006	2260	2581	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the RDA strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.33 [4.31]	0.34 [4.50]	0.32 [4.17]	0.28 [3.66]	0.26 [3.30]	0.23 [2.98]
Quintile	NYSE	EW	0.56 [8.39]	0.62 [9.54]	0.54 [8.86]	0.47 [7.65]	0.40 [6.87]	0.34 [5.96]
Quintile	Name	VW	0.30 [3.96]	0.32 [4.12]	0.30 [3.87]	0.28 [3.49]	0.25 [3.16]	0.24 [2.94]
Quintile	Cap	VW	0.25 [3.30]	0.26 [3.43]	0.24 [3.16]	0.20 [2.57]	0.18 [2.36]	0.15 [1.97]
Decile	NYSE	VW	0.34 [3.77]	0.33 [3.66]	0.29 [3.23]	0.23 [2.53]	0.25 [2.72]	0.20 [2.21]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.29 [3.83]	0.31 [4.05]	0.28 [3.71]	0.26 [3.44]	0.23 [3.07]	0.22 [2.90]
Quintile	NYSE	EW	0.37 [5.04]	0.45 [6.29]	0.37 [5.55]	0.33 [5.01]	0.23 [3.66]	0.20 [3.33]
Quintile	Name	VW	0.27 [3.48]	0.29 [3.74]	0.27 [3.47]	0.25 [3.27]	0.23 [2.95]	0.22 [2.83]
Quintile	Cap	VW	0.21 [2.83]	0.23 [2.97]	0.20 [2.69]	0.18 [2.37]	0.16 [2.14]	0.15 [1.93]
Decile	NYSE	VW	0.30 [3.32]	0.29 [3.19]	0.25 [2.78]	0.21 [2.41]	0.22 [2.41]	0.19 [2.15]

Table 3: Conditional sort on size and RDA

This table presents results for conditional double sorts on size and RDA. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on RDA. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high RDA and short stocks with low RDA. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	RDA Quintiles					RDA Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.43 [1.67]	0.80 [3.12]	0.87 [3.51]	0.96 [3.92]	1.05 [4.61]	0.65 [7.02]	0.70 [7.63]	0.64 [7.32]	0.60 [6.67]	0.47 [5.61]	0.44 [5.20]
	(2)	0.55 [2.40]	0.82 [3.48]	0.82 [3.48]	0.97 [4.32]	0.91 [4.31]	0.36 [4.31]	0.42 [4.99]	0.33 [4.16]	0.31 [3.79]	0.22 [2.74]	0.21 [2.58]
	(3)	0.50 [2.48]	0.68 [3.17]	0.80 [3.67]	0.79 [3.87]	0.97 [5.07]	0.48 [5.96]	0.51 [6.32]	0.46 [5.75]	0.43 [5.25]	0.37 [4.63]	0.35 [4.32]
	(4)	0.46 [2.49]	0.64 [3.17]	0.83 [4.11]	0.76 [4.08]	0.80 [4.41]	0.34 [4.28]	0.36 [4.52]	0.29 [3.82]	0.29 [3.70]	0.10 [1.42]	0.12 [1.61]
	(5)	0.47 [2.90]	0.48 [2.69]	0.50 [2.86]	0.58 [3.52]	0.68 [4.34]	0.22 [2.40]	0.22 [2.38]	0.21 [2.26]	0.17 [1.82]	0.18 [1.91]	0.15 [1.60]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	RDA Quintiles					RDA Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	374	375	373	372	366	31	33	38	29	28	
	(2)	108	106	107	107	105	56	55	56	55	55	
	(3)	78	77	77	77	77	97	95	97	99	99	
	(4)	65	65	65	65	65	202	207	213	211	217	
(5)	60	60	60	60	60	1452	1425	1680	1643	1946		

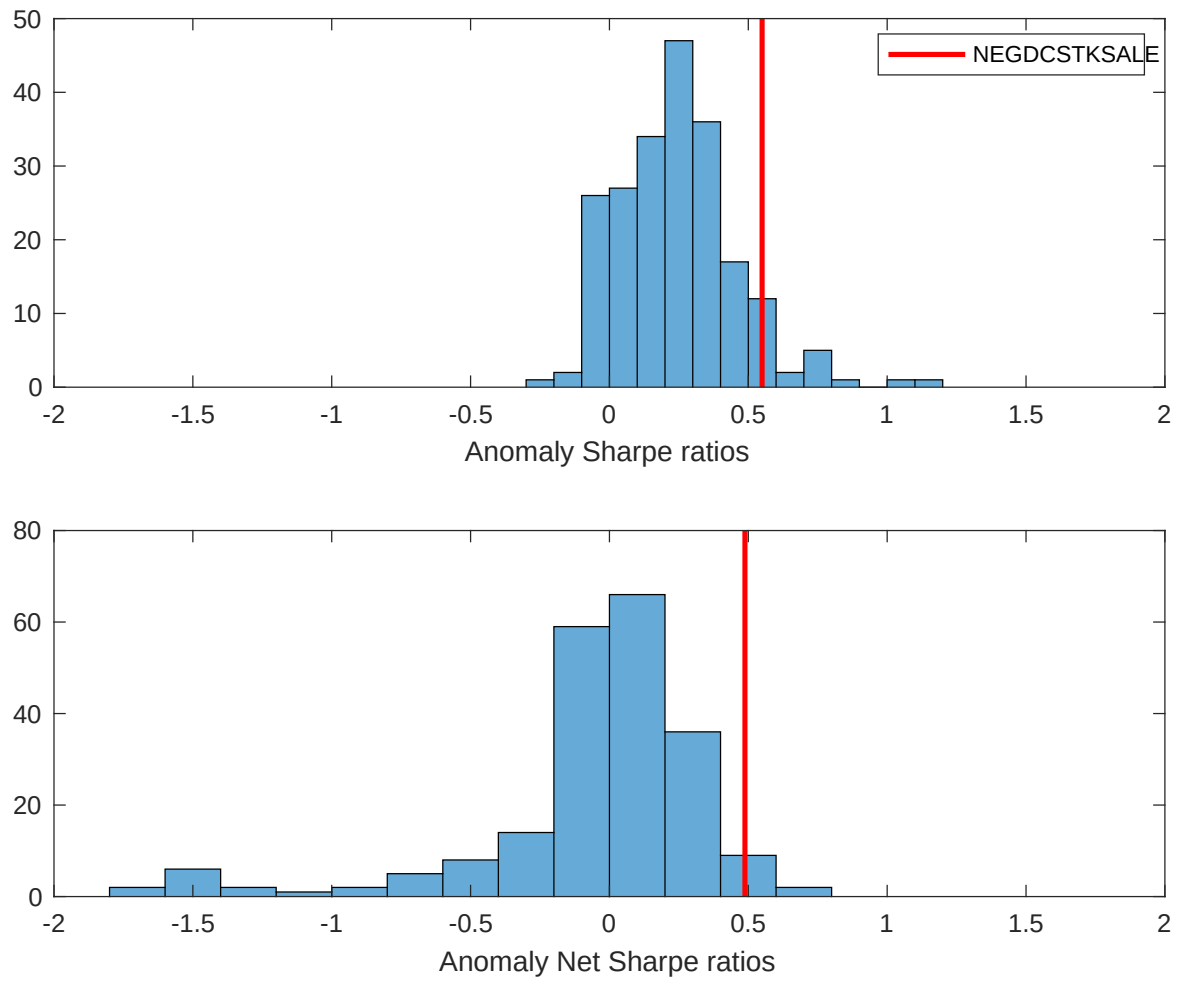


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the RDA with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

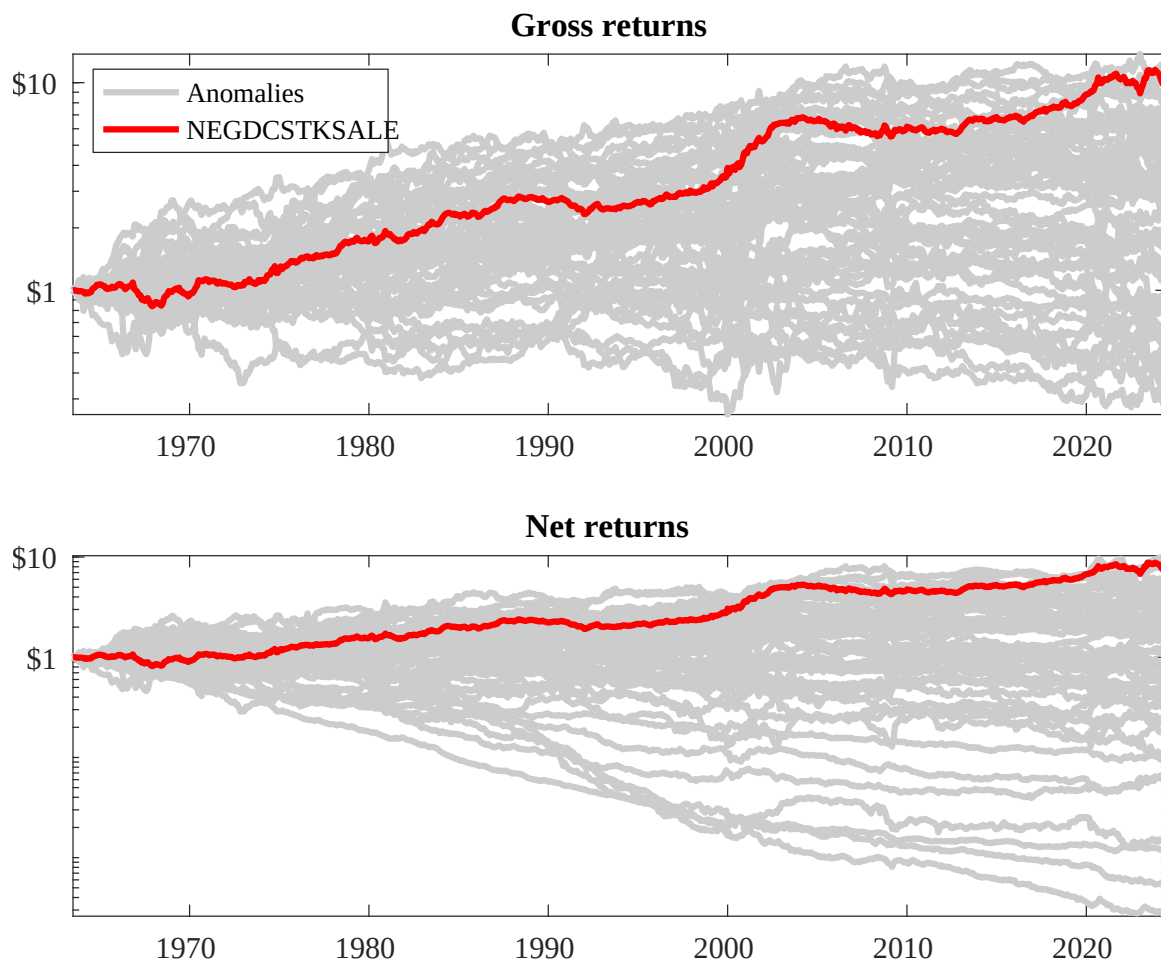
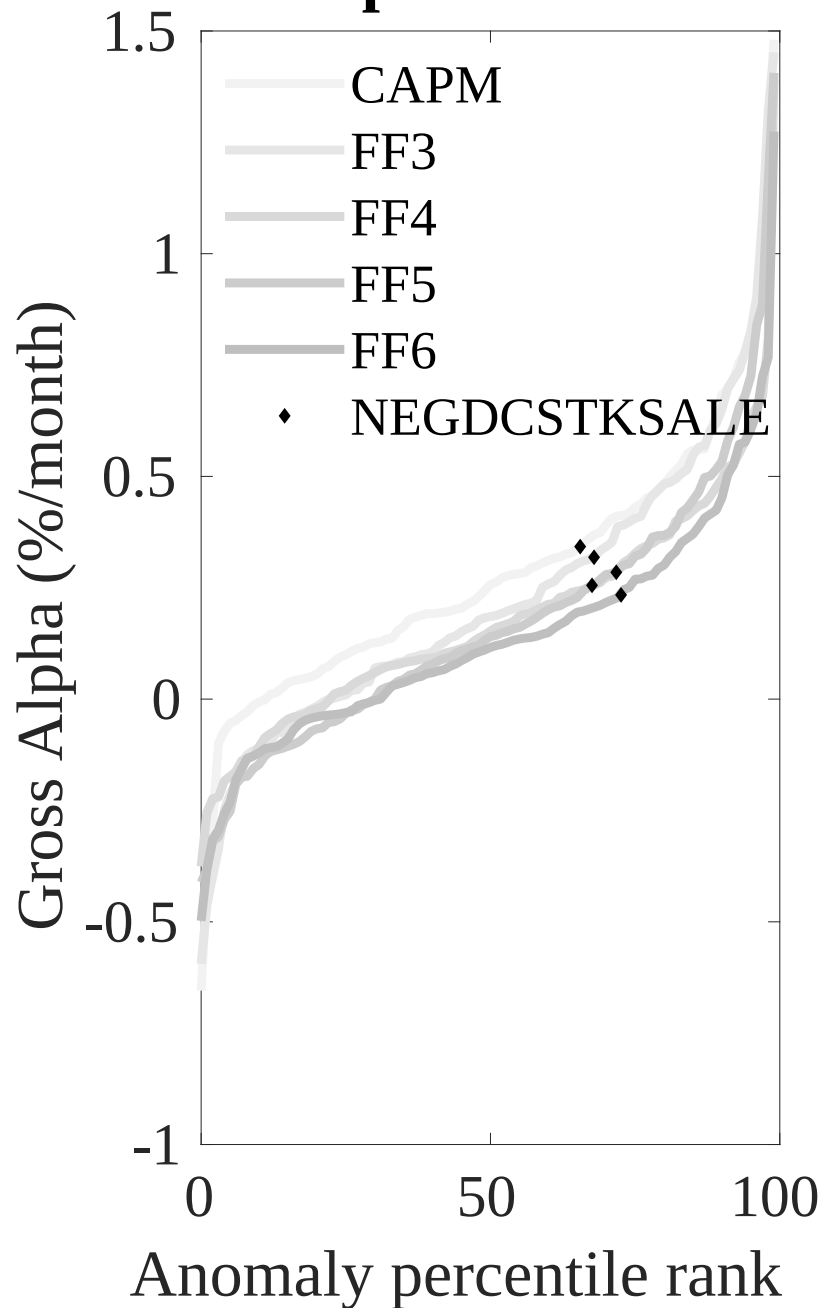


Figure 3: Dollar invested.

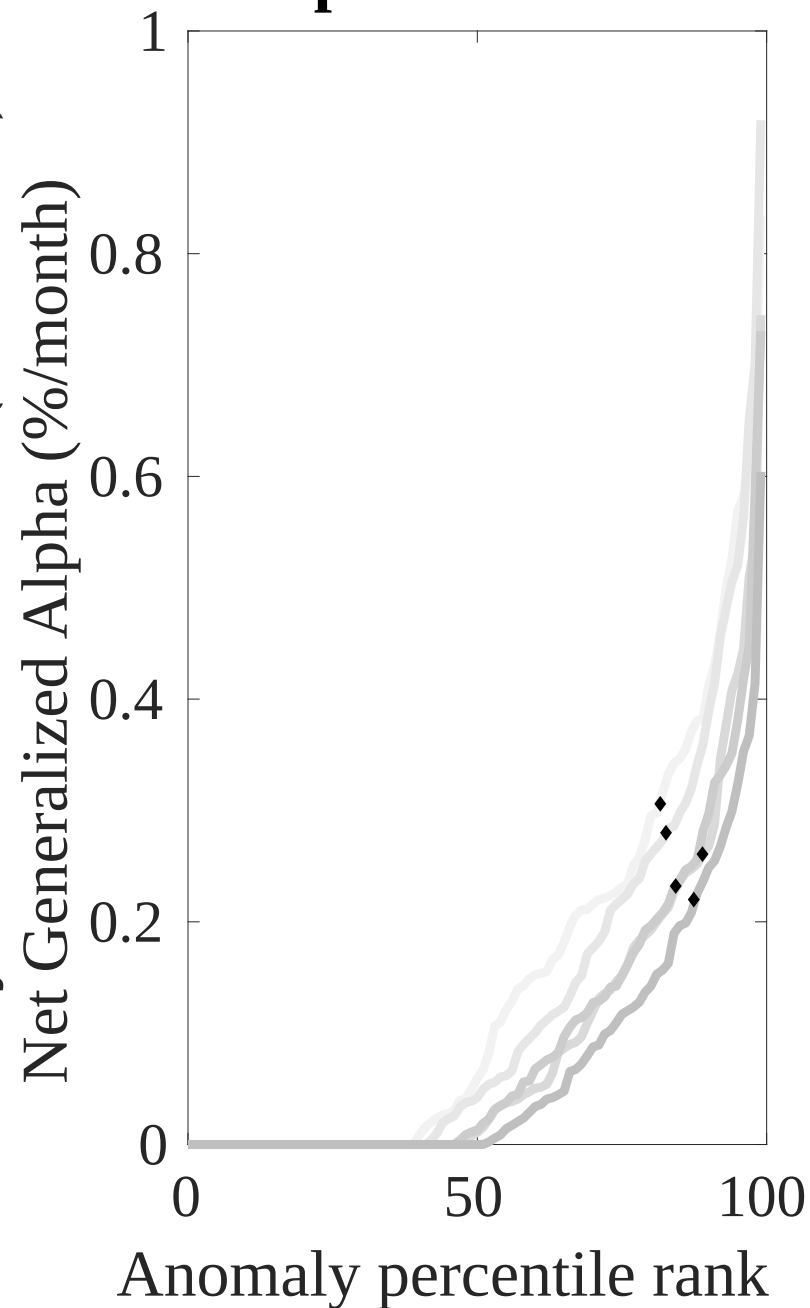
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the RDA trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles

Novy-Marx and Velikov (RFS, 2016)



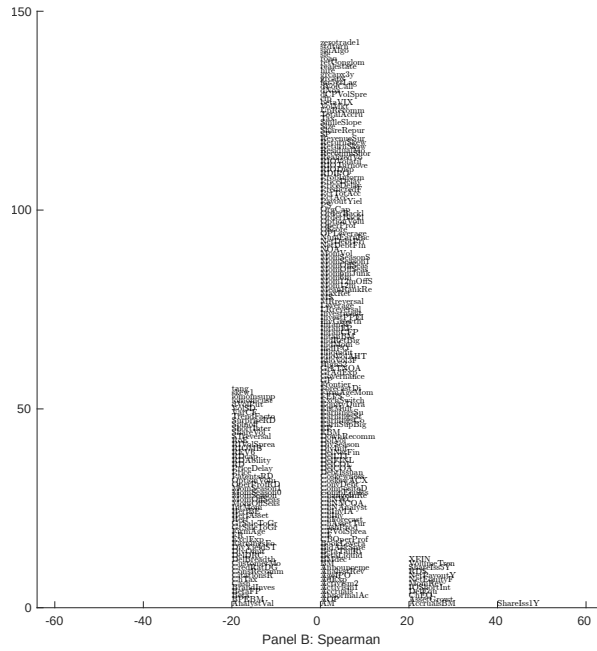
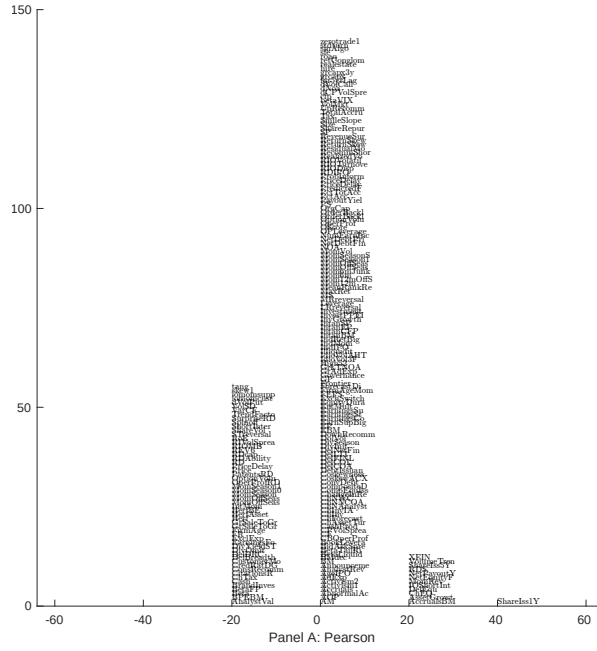


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with RDA. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

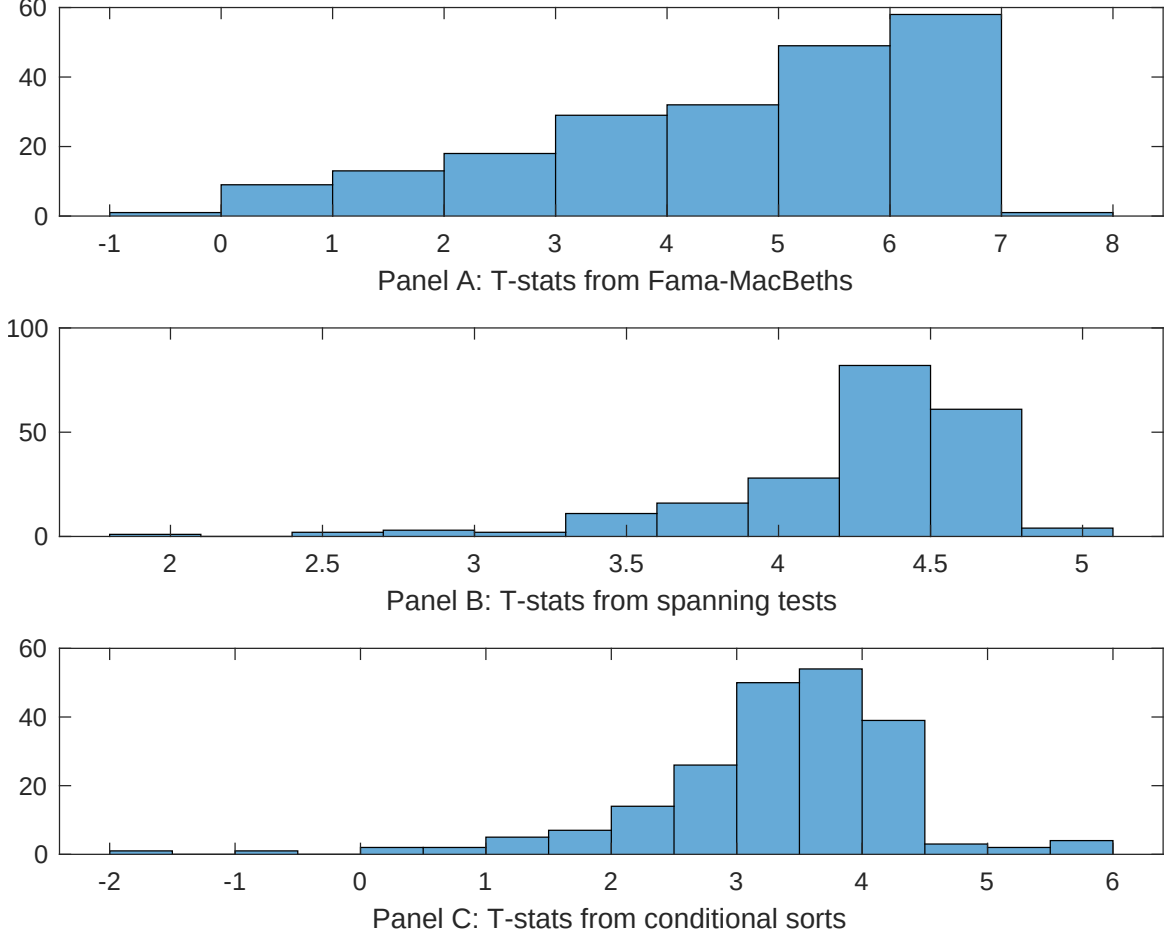


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of RDA conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{RDA} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{RDA} RDA_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{RDA,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on RDA. Stocks are finally grouped into five RDA portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted RDA trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on RDA. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{RDA}RDA_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Share issuance (1 year), Share issuance (5 year), Net Payout Yield, Growth in book equity, Change in equity to assets, Momentum and LT Reversal. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [6.11]	0.13 [6.47]	0.12 [5.44]	0.17 [6.93]	0.13 [5.81]	0.41 [1.20]	0.12 [2.82]
RDA	0.32 [4.73]	0.27 [3.58]	0.29 [3.64]	0.36 [5.14]	0.40 [5.72]	0.24 [1.14]	-0.24 [-0.86]
Anomaly 1	0.13 [5.90]						0.83 [1.57]
Anomaly 2		0.26 [4.81]					0.23 [2.05]
Anomaly 3			0.20 [1.74]				0.11 [1.18]
Anomaly 4				0.38 [3.13]			0.50 [2.06]
Anomaly 5					0.12 [3.14]		-0.15 [-1.84]
Anomaly 6						0.10 [4.18]	0.11 [3.66]
# months	715	715	715	720	720	647	501
$\bar{R}^2(\%)$	0	0	1	0	1	2	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the RDA trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{RDA} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Share issuance (1 year), Share issuance (5 year), Net Payout Yield, Growth in book equity, Change in equity to assets, Momentum and LT Reversal. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.22 [2.91]	0.16 [2.12]	0.26 [3.44]	0.22 [2.91]	0.23 [3.01]	0.21 [2.77]	0.15 [2.10]
Anomaly 1	26.05 [6.78]						7.82 [1.71]
Anomaly 2		29.96 [7.44]					18.52 [4.07]
Anomaly 3			16.12 [5.43]				6.94 [2.17]
Anomaly 4				30.23 [7.41]			31.51 [5.46]
Anomaly 5					10.71 [2.78]		-17.43 [-3.30]
Anomaly 6						4.81 [4.58]	3.95 [3.95]
mkt	2.06 [1.14]	2.76 [1.53]	3.92 [2.09]	2.41 [1.34]	1.22 [0.66]	1.57 [0.86]	4.90 [2.74]
smb	3.41 [1.30]	4.41 [1.69]	5.01 [1.86]	1.46 [0.56]	2.33 [0.87]	0.50 [0.19]	2.97 [1.15]
hml	-2.38 [-0.68]	0.65 [0.19]	-7.97 [-2.13]	-5.40 [-1.54]	-3.09 [-0.85]	-2.88 [-0.81]	-6.17 [-1.73]
rmw	-9.55 [-2.33]	-6.19 [-1.63]	-4.83 [-1.22]	5.96 [1.70]	5.51 [1.52]	7.03 [1.94]	-8.54 [-2.00]
cma	6.41 [1.14]	8.10 [1.49]	9.01 [1.57]	-6.62 [-1.03]	11.89 [1.82]	20.23 [3.87]	-11.63 [-1.79]
umd	2.00 [1.11]	1.23 [0.68]	4.47 [2.45]	2.67 [1.50]	3.47 [1.89]	-1.49 [-0.71]	-2.68 [-1.34]
# months	728	728	728	732	732	728	728
$\bar{R}^2(\%)$	10	11	8	11	5	6	18

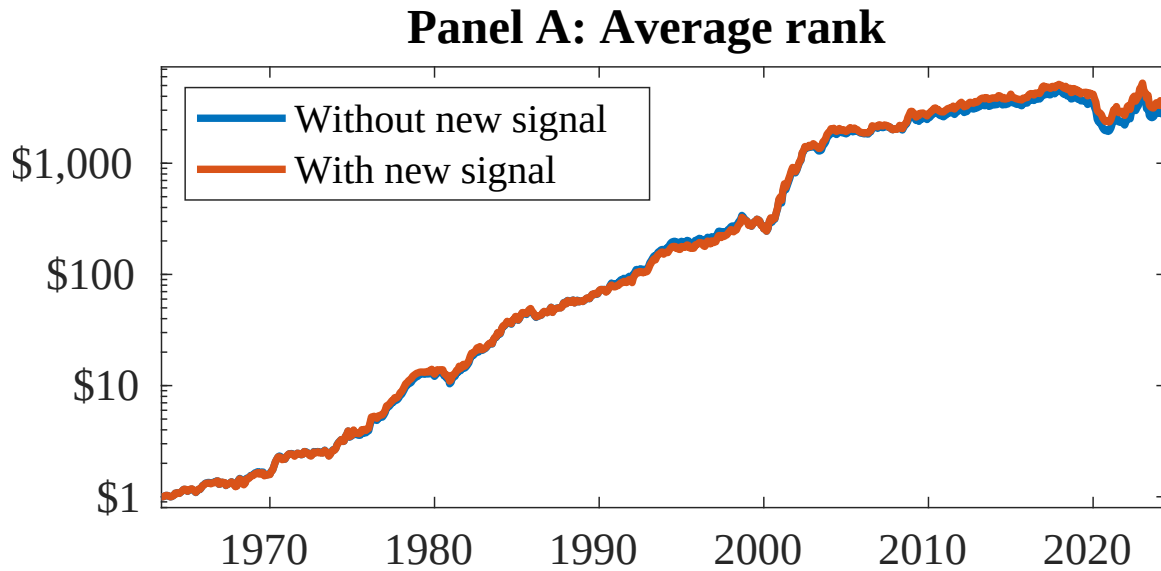


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as RDA. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

References

- Barro, R. J. (2006). Rare disasters and asset markets in the twentieth century. *Quarterly Journal of Economics*, 121(3):823–866.
- Bolton, P., Chen, H., and Wang, N. (2011). A unified theory of tobin’s q, corporate investment, financing, and risk management. *Journal of Finance*, 66(5):1545–1578.
- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.
- Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.
- Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.
- Cochrane, J. H. (1991). Production-based asset pricing and the link between stock returns and economic fluctuations. *Journal of Finance*, 46(1):209–237.
- Cooper, M. J., Gulen, H., and Schill, M. J. (2008). Asset growth and the cross-section of stock returns. *Journal of Finance*, 63(4):1609–1651.
- Daniel, K. and Titman, S. (2006). Market reactions to tangible and intangible information. *Journal of Finance*, 61(4):1605–1643.
- Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance*, Forthcoming.
- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.
- Fama, E. F. and French, K. R. (2008). Dissecting anomalies. *Journal of Finance*, 63(4):1653–1678.

- Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.
- Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.
- Gomes, J. and Schmid, L. (2010). Levered returns. *Journal of Finance*, 65(2):467–494.
- Liu, L. X., Whited, T. M., and Zhang, L. (2009). Investment-based expected stock returns. *Journal of Political Economy*, 117(6):1105–1139.
- Livdan, D., Sapriza, H., and Zhang, L. (2009). Credit market frictions and capital structure dynamics. *Journal of Financial Economics*, 93(2):155–174.
- Myers, S. C. and Majluf, N. S. (1984). Corporate financing and investment decisions when firms have information that investors do not have. *Journal of Financial Economics*, 13(2):187–221.
- Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2023). Assaying anomalies. *Working paper*.
- Pontiff, J. and Woodgate, A. (2008). Share issuance and cross-sectional returns. *Journal of Finance*, 63(2):921–945.
- Zhang, L. (2005). The value premium. *Journal of Finance*, 60(1):67–103.